

# Ryan J. Hamby

New York, NY | ryan.j.hamby@gmail.com

Portfolio: ryanjhamby.github.io | LinkedIn: linkedin.com/in/ryan-j-hamby | GitHub: github.com/RyanJHamby

Software Engineer at AWS building fault-tolerant, multi-tenant distributed systems, with merged open-source contributions to vLLM and Firecracker and independent low-latency systems work in Rust and C++20.

## Skills

**Languages:** Rust, C++ (C++20), Python, Java (strong) | Go, TypeScript, SQL (familiar)

**ML / GPU Systems:** vLLM (contributor), PyTorch Distributed (DDP), NCCL, NVML, KV Cache, CUDA Graphs

**Systems & Performance:** Concurrency, Lock-Free, Memory Ordering, Profiling (perf), ThreadSanitizer, Arrow

**Distributed Systems:** Kafka, gRPC/Protobuf, Exactly-Once, Idempotency, Circuit Breakers, Stream Processing

**Cloud & Infra:** AWS, Linux, Kubernetes, Docker, Terraform, Redis, PostgreSQL, Prometheus, CI/CD

## Experience

### Software Development Engineer II

Dec. 2024 – Present

*Amazon Web Services | AWS Marketplace Payments*

New York, NY

- Owned dual-partition, multi-tenant infrastructure upgrades with cross-partition DynamoDB custom-upload replication, enabling AWS Marketplace expansion into the EU sovereign cloud supporting \$112M+/yr in new GMV
- Led the build of the Payment Adapter Service (parallel third-party verification, circuit breakers, idempotent retries, async callbacks), cutting Japanese seller onboarding from 6 days to 5 minutes and unlocking \$10M+/yr in sales
- Built a Java event-driven sync service guaranteeing exactly-once delivery via SQS FIFO and DynamoDB conditional writes across 3 payment workflows, eliminating \$2M+/yr in reconciliation errors

### Software Development Engineer

Mar. 2023 – Nov. 2024

*Amazon Web Services | AWS IoT SiteWise, AWS Marketplace Payments*

Boston, MA / New York, NY

- Replaced a coarse synchronized block with a lock-stripped ConcurrentHashMap dedup index in the SiteWise Java control plane, reconciling multi-tenant composite-model name collisions and unblocking 10K+ concurrent writes
- Engineered IoT SiteWise ExternalId system: sparse GSI indexing, cross-account uniqueness validation, idempotent migration APIs, cutting customer onboarding time 40% and enabling 10K+ concurrent devices per account
- First engineer on the new AWS IoT SiteWise Boston team; onboarded 7 engineers to production in 30 days

## Open Source

- vLLM** (#48420, merged): Root-caused a Qwen3-Omni crash on audio-less video in the multimodal processor; fixed two follow-on false-positive paths in processor caching and parent/child processor inheritance
- Firecracker** (#6032, merged): Fixed VMM snapshot-restore panic from a masked `EventFd` fd-exhaustion error

## Projects

### GPU Flight Recorder | *Straggler detection and attribution for distributed GPU training*

Mar. 2026 – Present

- Built Go agents and a gRPC coordinator joining PyTorch NCCL collective timings with per-GPU NVML telemetry to find the straggling rank and rank likely causes (throttling, ECC, PCIe/NVLink, host stall)
- Replaced per-collective thresholds (0/30 detected at 10% jitter) with clock-skew-free block-median lag tests; 30/30 detection and attribution across 9 injected fault types with 0 false stragglers in simulation

### FlowState | *Rust temporal join engine for streaming ML feature pipelines*

Mar. 2026

- Built a Rust as-of join engine with  $O(n+m)$  merge-scan kernels, zero-copy Arrow via PyCapsule, and lock-free `UnsafeCell` writes across Rayon workers; 132 Rust tests incl. proptest checks
- Designed streaming incremental join with watermark-based emission, per-symbol state management, and configurable lateness tolerance for real-time point-in-time feature construction without look-ahead bias

### Low-Latency Order Book Engine | *C++20 matching engine benchmarked on EC2*

Sep. 2025 – Sep. 2026

- Engineered C++20 order matching engine with thread-local slab pools, cache-line-padded lock-free SPSC ingestion, and force-inlined matching; on EC2 `c6i.large`: P50 0.21  $\mu$ s, P99.9 3.1–3.2  $\mu$ s, up to 1.8M orders/sec

### Zonevex | *Founder, grid-interconnection alternative-data startup*

Mar. 2026 – Aug. 2026

- Built an async LLM-OCR pipeline extracting financing terms from scanned county records (PostGIS validation, Redis Streams workers); 86% out-of-sample accuracy against 350 real ERCOT interconnection outcomes

## Education

**University of Michigan** | B.S.E. in Computer Science, Minor in Electrical Engineering

Dec. 2022

GPA: 3.65, Magna Cum Laude | *Coursework:* Computer Organization, Signals & Systems, Semiconductors